2-4

convert hex to dec → 00 00 00 C4 =

a) $0x000000C4 = \underbrace{0000}\underbrace{0000}\ \underbrace{0000}\underbrace{0000}\ \underbrace{0000}\underbrace{0000}\ \underbrace{1100}\underbrace{0100}$

MSB = 0 = positive

$0 \cdot 16^7 + 0 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 4 \cdot 16^0 = 0 + 0 + 0 + 0 + 0 + 0 + 192 + 4 = 196$

b) $0xFFFFFEC6$ → FF FF FE C6 = 11111111 11111111 11111110 11000110

MSB = 1 = negative

00000000 00000000 00000001 00111001 +1

$= \underbrace{0000}\underbrace{0000}\ \underbrace{0000}\underbrace{0000}\ \underbrace{0000}\underbrace{0001}\ \underbrace{0011}\underbrace{1010}$

$= 0 \cdot 16^7 + 0 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 1 \cdot 16^2 + 3 \cdot 16^1 + 10 \cdot 16^0$

$= 256 + 48 + 10 = 314$

↳ -314

c) $0xFFFFFFFF$ = 11111111 11111111 11

MSB = 1 = neg. num

↳ 00000000 00000000 00000000 00000000 +1

= 00000000 00000000 00000000 00000001

$0 \cdot 16^7 + 0 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 0 \cdot 16^2 + 0 \cdot 16^1 + 1 \cdot 16^0$

$= 0 + 0 + 0 + 0 + 0 + 0 + 0 + 1 = 1$

$= -1$